

Filippo Dall'Asta

Embedded Software Engineer

 filippo.dallas@gmail.com

 +39 340 689 7354

 Milan, Italy

 [LinkedIn](#)

◆ profile

Embedded Software Engineer with experience in **Embedded Linux, firmware development**, and hardware/software integration. Comfortable across the stack: embedded C, customizing Linux systems with Buildroot, and designing Python services for connected devices. Hands-on with BLE, BlueZ, D-Bus, REST, MQTT, automated testing, and Linux system integration.

◆ experience

07/2024 – Present

Pirelli · Milan

Embedded Software Engineer

- Developed embedded firmware in C for connected devices, implementing new features and supporting automated validation and HIL testing.
- Designed and built a customized Embedded Linux OS with Buildroot, replacing an unsupported legacy environment and adapting the root filesystem, networking, packages, and system services to project requirements.
- Redesigned and rewrote a device-control service from scratch, replacing a monolithic serial AT-command architecture with a modular Python-based system.
- Built the service on Flask, Bleak, BlueZ and dbus-fast, translating factory REST commands into BLE operations including scanning, connection, characteristic read/write and device procedures.
- Implemented MQTT communication using Mosquitto and Paho-MQTT to publish operation results and device status back to factory systems.
- Integrated Linux service startup, health checks and device communication, improving maintainability and extensibility of the gateway software.

07/2023 – 07/2024

Cisco Photonics · Milan

Jr. Hardware Engineer

- Troubleshoot and repaired faulty PCB boards, supporting hardware development and validation.
- Performed FMEA and electrical measurements to identify design risks and verify compliance with technical specifications.
- Supported STM microcontroller debugging and hardware/software integration activities.

◆ education & skills

MSc Biomedical Engineering — Electronic Technologies

Politecnico di Milano · 2020–2023 · 110/110

BSc Biomedical Engineering

University of Bologna · 2017–2020 · 102/110

Computer Science · Erasmus exchange

Luleå University of Technology · 2022

programming

C Python Bash MATLAB

embedded / linux

Embedded Linux Buildroot BLE BlueZ D-Bus STM32 Silicon Labs CMake

networking / apis

REST Flask MQTT Mosquitto Paho-MQTT

testing / tools

Git CI/CD HIL Testing FMEA Oscilloscope Ellisys BT Analyzer

◆ personal projects

Personal Homelab

Proxmox VE · Linux VMs · LXC · Networking

Built and maintain a virtualized home infrastructure with hands-on experience in network configuration, VPN, DNS, reverse proxy, storage and service deployment.

Memorial Erika Reverberi — Website

Vercel · Supabase

Designed and built the official website for the Memorial Erika Reverberi sports tournament, deployed on Vercel with Supabase as the database backend.

[Visit site ↗](#)

◆ additional

languages

Italian, English, French, Spanish

volunteering

Cofounder & Organizer, Memorial Erika Reverberi — 2017–Present

certificate

Cambridge English C1 — 2020

Milan, Italy – last built 2026

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